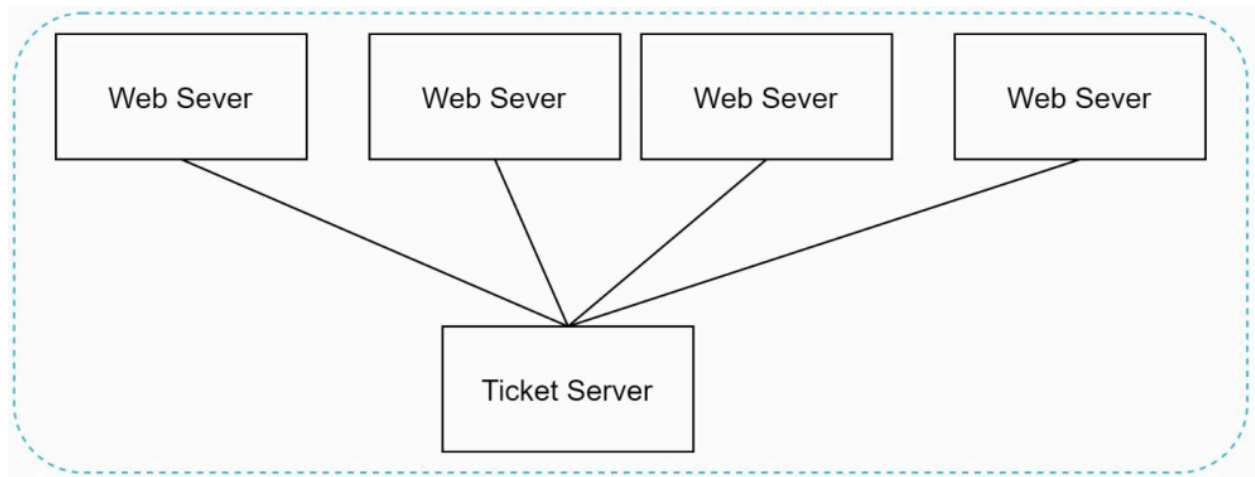


Ticket Server

Ticket servers are another interesting way to generate unique IDs in distributed systems. Flickr developed ticket servers to generate distributed primary keys.

How it works



The idea is to use a **centralized auto_increment feature** in a single database server (called the *Ticket Server*).

- Every request for a new ID goes to this central server
- The server increments a counter and returns a unique ID
- This ensures **global uniqueness** since only one source generates IDs

Pros

- Numeric IDs
- Easy to implement
- Works well for small to medium-scale systems

Cons

- **Single point of failure:** if the ticket server goes down, all dependent systems stop generating IDs
- To avoid this, multiple ticket servers can be introduced, but this leads to **synchronization challenges**